Code

```
clc;
clear;
close all;

% Transfer function
num = [1];
den = [1 2 5];
G = tf(num, den);

% Open-loop response
figure;
step(G);
title('Open Loop Response');

% PID Controller
Kp = 10;
Ki = 5;
Kd = 2;
C = pid(Kp, Ki, Kd);

% Closed-loop system
T = feedback(C*G, 1);

figure;
step(T);
title('Closed Loop Response with PID');
```

```
% Disturbance simulation
t = 0:0.01:10;
u = ones(size(t)); % step input
disturbance = zeros(size(t));
disturbance(t>=5) = -0.5; % disturbance at t=5

[y, t] = lsim(T, u + disturbance, t);

figure;
plot(t, y);
title('Response with Disturbance');
xlabel('Time (s)');
ylabel('Output');
```

Explanation

This project is based on modeling a control system using a transfer function in MATLAB. The system takes an input signal (step input) and produces an output response based on its dynamics.

Initially, a step input is applied to observe how the system reacts over time. The response graph helps in understanding key characteristics like rise time, settling time, and stability.

Next, a feedback loop is introduced using a summation block. This allows the system to compare the desired input with the actual output and adjust accordingly, improving accuracy.

To make the system more realistic, a disturbance input is added. This represents external factors that can affect system performance. By analyzing the output under disturbance, we can evaluate how robust and stable the system is.

My Idea Behind the Project

My idea was to create a simple but effective control system model that clearly demonstrates how real-world systems behave. Instead of making it too complex, I focused on understanding the fundamentals like system response, feedback, and disturbance handling.

The goal was to visually show how a system changes when disturbances are introduced and how feedback helps in correcting the output.